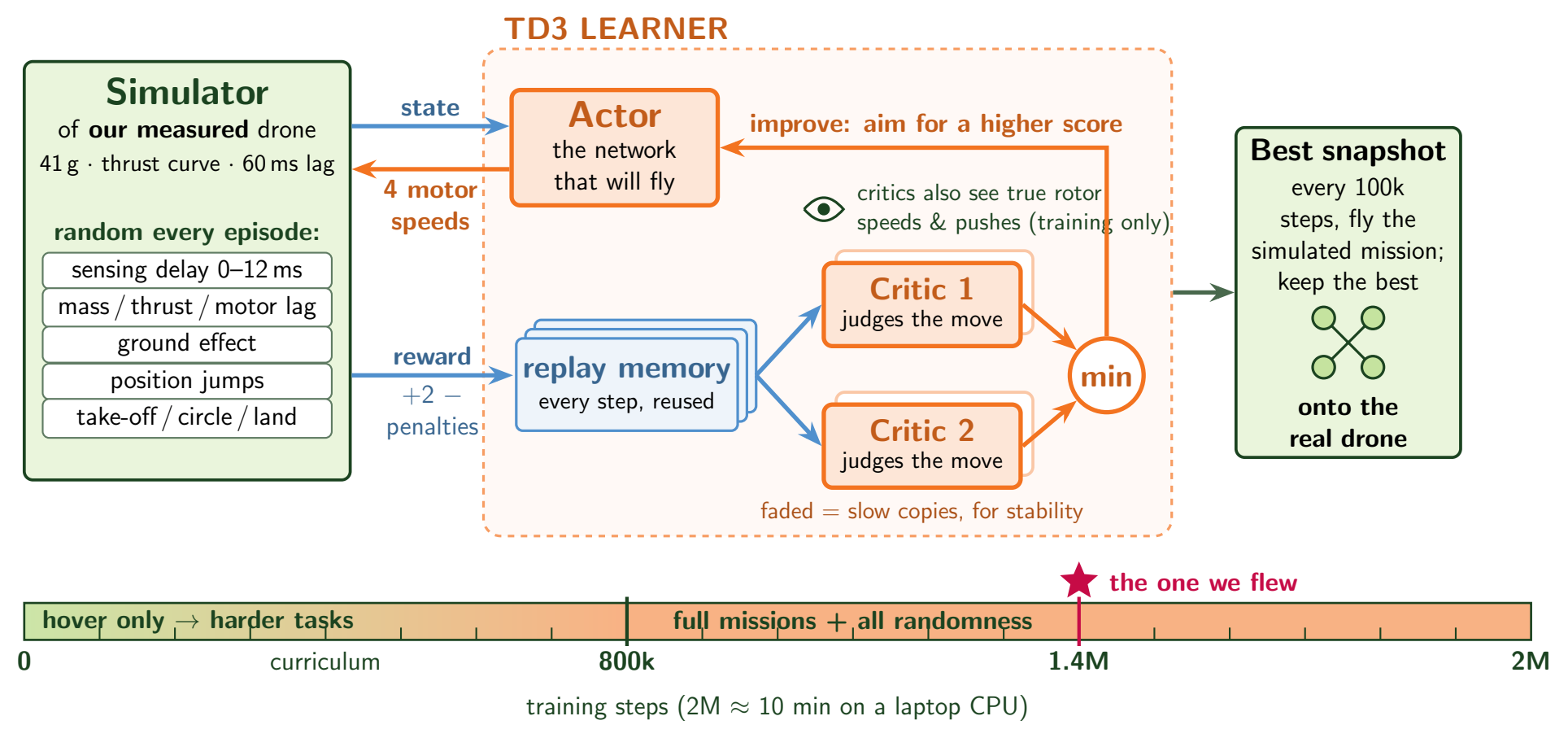
TD3 LEARNER
Simulator
of our measured drone
41 g · thrust curve · 60 ms lag
random every episode:
sensing delay 0–12 ms
mass / thrust / motor lag
ground effect
position jumps
take-off / circle / land
state
4 motor speeds
reward
+2 − penalties
Actor
the network that will fly
improve: aim for a higher score
critics also see true rotor speeds & pushes (training only)
replay memory
every step, reused
Critic 1
judges the move
Critic 2
judges the move
min
faded = slow copies, for stability
Best snapshot
every 100k steps, fly the simulated mission; keep the best
onto the real drone
the one we flew
hover only → harder tasks
full missions + all randomness
0
curriculum
800k
1.4M
2M
training steps (2M ≈ 10 min on a laptop CPU)